Isaac Newton

British scientist Isaac Newton was born in the village of Woolsthorpe, England, in 1642. One of the leading minds of the 17th-century scientific revolution, he is best known for outlining the law of universal gravitation to explain what holds the Universe together.

Schoolboy and student

Newton's interest in science and mechanics became apparent at an early age. An uncle recognized his ability and encouraged him to continue his studies at a university. In 1661, he became a student at Trinity College, Cambridge, England.

Escape from the plague

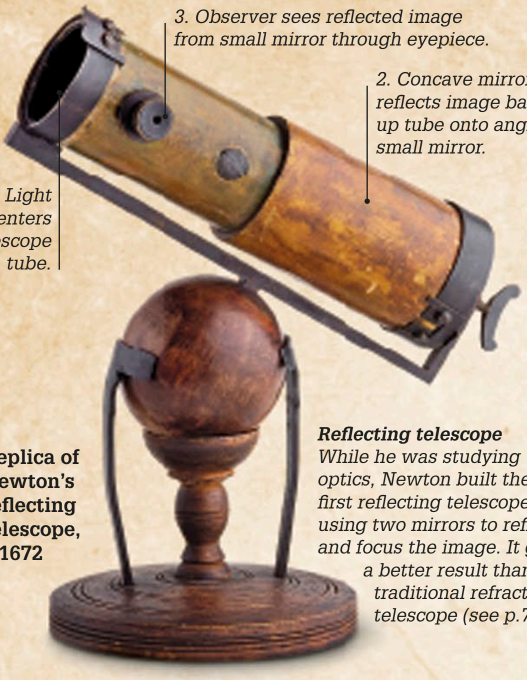
When the plague broke out in Cambridge in 1665, Newton withdrew to Woolsthorpe. He is said to have developed his theory of gravity after seeing an apple fall from a tree in the orchard there. This is probably just a story, but it was during his time at Woolsthorpe that he developed his ideas on gravitation and made his first experiments with light.

Cambridge professor

Returning to Cambridge, Newton was appointed Lucasian Professor of Mathematics at the age of 26. In 1687, he published *Philosophiae Naturalis Principia Mathematica* (usually called the *Principia Mathematica)*, one of the most important works in the history of science, and where he described his three laws of motion.

Later years

In 1689, Newton became a Member of Parliament and moved to London. Appointed Master of the Royal Mint in 1699, he reformed the coinage and took severe measures against forgers. He was elected President of the Royal Society in 1703 and made a knight in 1705. He died in 1727 and was buried in Westminster Abbey.



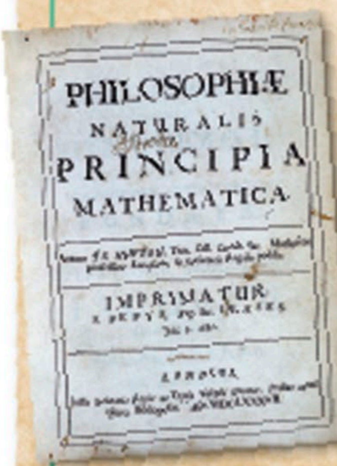
Replica of Newton's reflecting telescope, c 1672

Reflecting telescope
While he was studying optics, Newton built the first reflecting telescope, using two mirrors to reflect and focus the image. It gave a better result than the traditional refracting telescope (see p.71).

> **"If I have seen further it is by standing on the shoulders of giants."**
>
> **Isaac Newton, in a letter to Robert Hooke, 1675, supposedly acknowledging earlier work by other scientists**

Principia Mathematica

Newton's most famous book was published with the help of fellow scientist Edmond Halley. In this book, Newton described the universal law of gravitation (see p.87) and the three laws of motion (below).


PHILOSOPHIÆ
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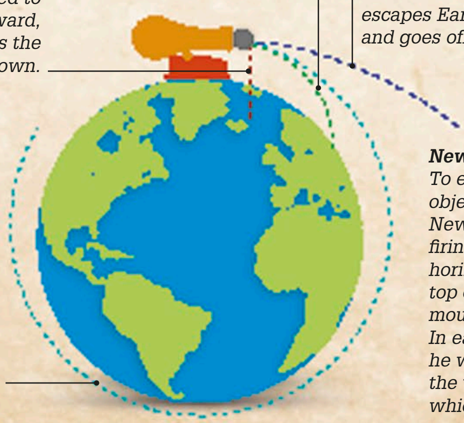
The laws of motion
1. An object remains at rest or continues moving in a straight line unless a force acts upon it.

2. The greater the mass of an object, the more force it will take to accelerate it.

3. For every action, there is an equal and opposite reaction.

Title page of *Principia Mathematica*



Newton's cannonball
To explain why one object orbits another, Newton imagined firing a cannonball horizontally from the top of a very high mountain on Earth. In each experiment he would increase the velocity (speed) at which the ball travels.